

Developer

PROFILE SUMMARY

- Senior Blockchain Developer with 8 years of experience shipping production smart contracts and protocol infrastructure on Ethereum and Polygon across DeFi lending and DEX protocols, on-chain identity, and L2 rollup infrastructure, specializing in gas-optimized Solidity, formal verification with Certora, and EIP-4844 rollup data work.
- Hands-on coverage across contract language (Solidity 0.8.x), contract framework (Foundry with OpenZeppelin Contracts), chain (Ethereum mainnet with Polygon PoS), and node client (Geth with Lighthouse), with scaling grounded in Optimism Bedrock with Arbitrum Nitro and credentialed as Cyfrin Updraft Security Researcher.
- Deep expertise in EVM contract architecture with upgradeable proxies and diamond patterns, rollup data with EIP-4844 blob carriers and fraud-proof design, validator infrastructure with slashing and MEV-aware block building, and tokenomics with ERC-20, ERC-721, and ERC-1155 standards and governance hooks, applying methodologies such as invariant fuzzing with Echidna and Foundry plus formal verification on critical paths and threat modeling against reentrancy, oracle manipulation, and flash-loan attacks to deliver audited, gas-efficient protocols that hold up under mainnet load and adversarial review.
- Engaged collaborator working cross-functionally with protocol research, security, infrastructure, and external audit teams in audit-heavy protocol engineering teams shipping to mainnet, contributing to ERC standard discussions, mainnet incident response, and security review boards with an ownership-first mindset and clean handoffs.
- Mentor who shares technical excellence and fosters a culture of audited, formally verified smart contracts and reproducible build and deploy practice through PR reviews and threat-model docs, while running the internal security review board and ERC standards working group and authoring widely cited OpenZeppelin Contracts modules and rollup tooling.

TECHNICAL SKILLS

Languages:	Solidity 0.8.x, Vyper, Rust with Anchor, Move, Yul, Huff, TypeScript, Go, Python
Contract Frameworks & Standards:	Foundry with OpenZeppelin Contracts, ERC-20, ERC-721, ERC-1155, EIP-2535 Diamonds, OpenZeppelin Contracts, Solady, upgradeable proxies, access control patterns
Chains & Protocols:	Ethereum mainnet with Polygon PoS, Ethereum mainnet, Solana, Polygon PoS and zkEVM, Cosmos SDK, Aptos, Sui, Avalanche subnets, Polkadot parachains
Node Clients & Networking:	Geth with Lighthouse, Geth, Erigon, Reth, Lighthouse, Prysm, validator key management, P2P gossip, libp2p, JSON-RPC, WebSocket, IPC
Security & Cryptography:	Slither, Mythril, Echidna, Foundry invariant tests, Certora formal verification, reentrancy, oracle manipulation, flash-loan vectors, MEV-aware design, EIP-712 signatures
L2 & Scaling:	Optimism Bedrock with Arbitrum Nitro, Optimism Bedrock, Arbitrum Nitro, zkSync Era, Polygon zkEVM, Linea, Scroll, StarkNet Cairo, OP Stack rollups, fraud proofs, validity proofs
Testing & Tooling:	Foundry (Forge, Cast, Anvil), Hardhat, Truffle, Echidna fuzzing, Slither static analysis, Tenderly simulations, Hardhat coverage, mainnet forking, gas snapshots
Certifications & Communities:	Cyfrin Updraft Security Researcher, Cyfrin Updraft Security Researcher, Secureum Bootcamp, ConsenSys Diligence trainings, OpenZeppelin Defender, Code4rena and Sherlock contests, ERC standards authorship

EDUCATION

Czech Technical University in Prague **M.Sc. in Cryptography and Computer**

Prague, Czechia • Oct 2013 - Jun

WORK EXPERIENCE

OpenZeppelin Senior Blockchain Developer

Prague, Czechia • Apr 2021 - Present

- Owned protocol and consensus work end to end on the **DeFi protocol portfolio across EVM chains securing \$4.2B in protocol TVL**, driving **consensus tuning, contract architecture, and security review** across **14 chains** with finality, block-production, and slashing rules tuned against mainnet load.
- Wrote and shipped smart contracts on **Solidity with EIP-2535 Diamonds, OpenZeppelin upgradeable proxies, and Solady for hot paths**, reusable Solady mixins on hot paths, role-gated upgrade flows, and a custom Vyper module on precompile-heavy contracts, taking **187 audited contracts** to mainnet that today secure **\$3.1B** in user funds without a critical post-deploy regression.
- Ran node infrastructure and P2P networking on **Geth and Erigon execution clients paired with Lighthouse and Prysm consensus clients across three regions**, libp2p gossip tuning on peer scoring, JSON-RPC and WebSocket endpoints behind a load-balanced edge, and an MEV-Boost relay setup across **64 validators**, holding consensus uptime at **99.94%** across the last 12 months without a missed attestation window.
- Hardened cryptography and contract security with **Slither and Mythril static analysis, Echidna invariant fuzzing, and Certora formal proofs on lending and bridge contracts**, ECDSA and EdDSA signature paths reviewed end to end, Merkle proof verifiers tightened against malleability, and zk-SNARK circuits audited under a structured threat model, disclosing **142 findings** to client teams and catching **11 critical bugs** before mainnet deploy.
- Scaled the L2 footprint with **Optimism Bedrock and Arbitrum Nitro rollup deployments with EIP-4844 blob carriers and a custom Polygon zkEVM bridge**, fraud-proof window analysis on the optimistic path, validity-proof batching on the zk path, and a state-channel exit hatch for high-frequency users, lifting rollup throughput from **18 TPS to 240 TPS** and onboarding **2.4M L2 wallets** across the rollup set.
- Designed on-chain architecture and tokenomics around **ERC-20 governance tokens with ERC-721 reputation NFTs, ERC-1155 reward batches, and on-chain treasury via OpenZeppelin Governor**, account-based state model on EVM and a UTXO-style accumulator on the bridge accounting side, with incentive curves calibrated against simulated voter behavior, shipping **38 executed DAO proposals** across **320k active token holders** without a quorum failure.
- Drove performance and gas optimization through **custom storage packing, transient storage with EIP-1153, Solady assembly snippets, and call data compression on the L2 path**, function-selector ordering, calldata-first ABI redesign, and warm-storage prefetch on hot read paths, cutting average swap gas from **218k gas down to 74k gas** across the protocol set and clearing **22 external audit engagements** with zero unresolved high-severity items.

Consensys Blockchain Engineer (MetaMask Snaps)

Berlin, Germany • Jul 2018 - Mar 2021

- Built backend services and on-chain integrations around **ethers.js and viem clients, Chainlink price feeds, and The Graph subgraphs for event indexing**, transaction signing flows wired through HSM-backed key custody, event listeners on Pub/Sub fan-out, and Eventarc retries on dropped logs, indexing **1.8M daily transactions** across **26 subgraphs**.
- Owned testing, auditing, and simulation with **Foundry forge with mainnet forking, Hardhat coverage, Tenderly simulations, and Echidna invariants on the wallet contracts**, attack-vector replays from past public exploits, gas-snapshot diffs on every PR, and a fuzzing corpus seeded from production traffic, raising contract coverage to **92%** and shipping clean reports across **9 external audit engagements**.
- Wired the wallet into decentralized systems through **IPFS pinning with Pinata, Arweave permanence, ENS resolvers, and W3C Decentralized Identifiers tied to a Ceramic profile graph**, Snap permission scopes locked to DID-issued credentials, and Lit Protocol session keys for delegated signing, reaching **540k wallets** across **18 DAO integrations**.
- Stood up multi-chain governance through **Snapshot off-chain voting tied to on-chain execution via OpenZeppelin Governor and Gnosis Safe multisigs**, LayerZero messages for cross-chain proposal relay, Axelar gateways for asset hooks, and timelock controllers gated on Safe approvals, executing **63 proposals** across **7 chains** with no signature replay incident.